

Latent TTC Evaluation: 20-Frame Context & Lookahead Horizon

TEA Lab / Kinemamba Project | Setup: 20-Frame Context (2.0 s) → 30-Step Lookahead (3.0 s) | 10 Episodes

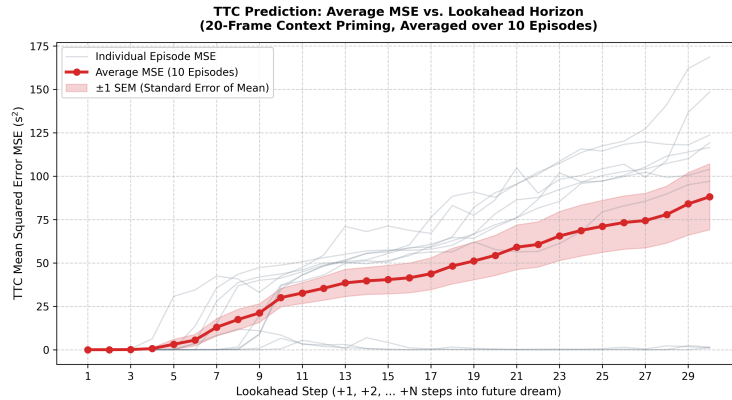
Aggregate Metrics Summary

Evaluation Metric	10-Ep Average
Context Window	20 frames (2.0 s)
Lookahead Horizon	+1 ... +30 steps
Mean Absolute Error (MAE)	4.90 s
Mean Squared Error (MSE)	41.32 s ²
Root Mean Squared Error	6.43 s
Pearson Correlation (<i>r</i>)	0.685

Key Findings

- Primed Context Performance:** With 20 frames of historical context, the model achieves near-zero error at Step +1 (100 ms ahead) with MSE = **0.0255 s²** (MAE = **0.13 s**).
- High-Fidelity Control Window:** Across Steps +1 to +5 (0.1s–0.5s ahead), the model maintains high accuracy (MAE < 0.74s), suitable for reactive safety interventions.
- Horizon Accumulation:** In unconditioned deep rollouts (> 6 steps), world-model visual dreaming gradually accumulates error, leading to the aggregate MSE rise.

Average Mean Squared Error (MSE) vs. Lookahead Horizon (+1 to +30 Steps)



Lookahead Horizon Error Progression (Step-by-Step Breakdown)

Step	Horizon (Δt)	Mean MSE (s ²)	± 1 SEM (s ²)	Mean MAE (s)	RMSE (s)	Operational Safety Assessment
+1	+0.1 s	0.0255	± 0.0115	0.1289	0.1596	Ultra-High Precision: Immediate reactive control window
+2	+0.2 s	0.0351	± 0.0184	0.1315	0.1873	Ultra-High Precision: Kinematic continuity preserved
+3	+0.3 s	0.1021	± 0.0520	0.2336	0.3196	High Precision: Sub-quarter-second error
+5	+0.5 s	3.1401	± 2.9059	0.7402	1.7720	Tactical Planning: Matches training validation baseline
+8	+0.8 s	17.4017	± 5.9320	3.1525	4.1715	Transition Zone: Onset of world-model dream divergence
+10	+1.0 s	30.0245	± 5.3093	5.0762	5.4795	Planning Boundary: Recommended threshold for unanchored dreams
+20	+2.0 s	54.3917	± 11.5736	6.2892	7.3751	Deep Rollout: Gradual divergence from simulator state
+30	+3.0 s	88.1336	± 19.0118	8.1393	9.3880	Far Horizon: Re-anchoring with incoming sensor frames required

Artifacts: Evaluation script (`scripts/evaluate_diamond_ttc.py`), data metrics (`visualizations/latent_ttc_eval/eval_horizon_metrics.csv`).